PrivIMU — Re-identification Risk in Anonymous Motion Sensor Traces

A reproducible IoT privacy study on the MotionSense dataset, with attack metrics, defence sweep and a Streamlit lab.

Group of 5 — Doctoral Training Programme, FF6 IoT module

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Repository : github.com/akiroussama/PrivIMU · **Notebook** : Google Colab · **Demo** : Streamlit (see §7)

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Abstract

Motion data are routinely collected by smartphones, wearables and connected devices through accelerometer and gyroscope sensors. Because they carry no name, photo or e-mail, they are often perceived as *low-risk telemetry* and shared with third-party services with little gatekeeping. **PrivIMU** challenges that assumption by asking whether such anonymous-looking traces still carry enough behavioural signature to **re-identify** a user from a known candidate set.

We build an end-to-end reproducible pipeline on the public MotionSense corpus (24 subjects, 6 activities, iPhone DeviceMotion at 50 Hz) and train a Random Forest classifier on 60 hand-crafted features per 1-second window. With a **5-fold GroupKFold** evaluation grouped by activity-trial — a split that prevents within-trial leakage between train and test — the classifier reaches **top-1 accuracy 39.06 %** and **top-3 accuracy 52.80 %**, against a random baseline of 4.17 % (*9.4× lift*). The mean posterior entropy leakage is $\Delta H = \log_2(24) - H(p) \approx 0.700$ bit, and per-window inference takes ≈ 42 ms on a CPU.

To make the result tangible, we ship a **Streamlit privacy lab** that ranks top-3 candidate identities for any uploaded CSV, runs a Gaussian-noise defence sweep, and exports an evidence card (JSON + Markdown). All numerical claims in this report are generated by code into `reports/metrics.json`, which is the single source of truth for the slides and the demo.

Scope note. *The 39 % top-1 figure is not a real-world identification system : it is a proof that the signal carries identity on a controlled 24-subject closed-set benchmark. Under larger populations, deeper models or device-fingerprint cross-correlation, the number can only grow. The right reaction is to treat motion data as personal data — not to wait for the attack at scale.*

ACRONYMS

HAR	Human Activity Recognition	IMU	Inertial Measurement Unit
IoT	Internet of Things	RF	Random Forest
OOF	Out-Of-Fold (predictions)	PII	Personally Identifiable Information
DP	Differential Privacy	FL	Federated Learning
ΔH	Leakage : $\log_2(K) - H(\mathbf{p})$ in bits	$H(\mathbf{p})$	Shannon entropy of posterior $\mathbf{p}$

39.06	52.80	0.700	42.0
% TOP-1	% TOP-3	ΔH BITS	MS / WINDOW
vs 4.17 % random — 9.4× lift	true subject in top 3 candidates	mean posterior leakage per window	CPU inference latency, RF 200 trees

1. Introduction and motivation

The Internet of Things has normalised the continuous capture of inertial signals. Every smartphone exposes accelerometer and gyroscope streams through the platform DeviceMotion API ; smartwatches and fitness bands sample the same channels at 50–200 Hz around the clock ; connected vehicles, e-scooters, AR/VR headsets and rehabilitation devices forward similar telemetry to back-end services. By default, this data is treated as *"low-risk"* — it carries no face, no voice, no name — and is shared across analytics SDKs with little gatekeeping.

This report studies whether that assumption holds. The scientific question is concise :

> **RESEARCH QUESTION**
>
> Can a supervised classifier trained on accelerometer + gyroscope windows **re-identify a user** among a closed candidate set, when the input contains no direct identifier ?

The hypothesis is that 1-second windows of six IMU channels carry enough **behavioural signature** — gait, posture micro-tremor, hand dynamics — to act as a quasi-identifier even after stripping every explicit identity field. The contributions of this work are :

1. **C1 — Reproducible pipeline.** An end-to-end Python package (`privimu.{data,features,model_rf,train,evaluate}`) covering dataset ingestion, windowing, feature extraction, training and reporting, with 8 unit tests and a CI workflow.

2. **C2 — Honest measurement.** Top-1 / top-3 accuracy, macro-F1, per-subject F1, posterior entropy and entropy leakage, all generated into `reports/metrics.json` with a **group-aware cross-validation** that explicitly forbids within-trial leakage between train and test.
3. **C3 — Interactive privacy lab.** A Streamlit application that, given any MotionSense-like CSV, ranks the most likely subjects, plots an identity-lock replay, runs a Gaussian-noise defence sweep and exports a downloadable evidence card.
4. **C4 — Privacy-by-design discussion.** A practical mapping from the measured leakage to the defence stack — data minimisation, on-device inference, federated learning, differential privacy — plus an explicit ethical scope and a list of limits.

2. Background and related work

Three research lines intersect at PrivIMU : human activity recognition (HAR), behavioural biometrics and sensor privacy leakage.

2.1 *Human activity recognition with IMU*

HAR studies use the same IMU channels as PrivIMU, but the prediction target is the *activity* rather than the *subject*. The benchmarks of Anguita *et al.* [1] (UCI HAR) and Malekzadeh *et al.* [2] (MotionSense) demonstrated that simple feature pipelines and shallow classifiers reach 90 %+ accuracy at recognising six daily activities. PrivIMU re-uses that feature engineering recipe but re-targets the label space.

2.2 *Gait and behavioural biometrics*

Gait recognition is a long-standing biometric problem ; surveys by Gafurov [3] and Hadid *et al.* [4] document a body of work from floor sensors, vision and dedicated motion units. Smartphone-based gait biometrics [5, 6] showed that accelerometer-only or accelerometer+gyroscope identification is feasible with cooperative protocols (controlled walking, fixed device placement). PrivIMU departs from these works by treating identification as a *privacy threat* rather than an *authentication tool*.

2.3 *Sensor privacy leakage*

A separate line of work shows that motion sensors, when accessed silently, leak more than expected : Mehrnezhad *et al.* [7] inferred touch input from web-page motion events ; Das *et al.* [8] fingerprinted smartphones via accelerometer manufacturing imperfections. PrivIMU sits between the HAR literature and these leakage studies : same dataset family as HAR, same threat-model spirit as leakage.

2.4 *Defences in the literature*

The mitigation toolbox combines **data minimisation** (lower sample rate, drop channels), **on-device computation** (avoid sending raw windows), **federated learning** [9] (train without centralising raw traces), **differential privacy** [10] (calibrated noise on aggregates) and feature-level sharing (release coarse statistics rather than samples). PrivIMU prototypes the simplest one — additive Gaussian noise on raw signals — as a pedagogical baseline and discusses the rest in §8.

Work	Sensors	Task	Subjects	Privacy framing
Anguita 2013 [1]	Accel + Gyro	Activity (HAR)	30	None
Malekzadeh 2018 [2]	Accel + Gyro (MotionSense)	Activity / privacy filter	24	Sensitive-attribute hiding
Derawi 2010 [5]	Accel	Identification (gait)	51	None (auth)
Mehrnezhad 2018 [7]	Web motion API	Touch inference	Web users	Side channel
Das 2016 [8]	Accel	Device fingerprint	107 phones	Tracking
PrivIMU (this work)	**Accel + Gyro (MotionSense)**	**Re-identification**	**24**	**Privacy-by-design lab**

Tab. 1 — Selected positioning of PrivIMU. The originality is the explicit *privacy threat* framing of an HAR-style pipeline, paired with an interactive defence demonstrator.

3. Dataset and preprocessing

We use the public **MotionSense** dataset of Malekzadeh *et al.* [2]. It contains 24 subjects performing six activities (walking, jogging, upstairs, downstairs, sitting, standing) while carrying an iPhone 6s in the front pocket. The DeviceMotion API exposes the high-level signals `attitude.{roll,pitch,yaw}`, `gravity.{x,y,z}`, `rotationRate.{x,y,z}` and `userAcceleration.{x,y,z}` at ≈ 50 Hz. PrivIMU keeps the six rotation-rate and user-acceleration channels — the gravity-corrected motion components — and ignores the `attitude` Euler angles, which encode an absolute orientation that varies with phone placement.

Property	Value
Dataset	MotionSense (DeviceMotion subset)
Subjects K	24
Activities	`wlk`, `jog`, `ups`, `dws`, `sit`, `std`
Sampling rate	~50 Hz
IMU channels used	6 (rotationRate.{x,y,z} + userAcceleration.{x,y,z})
Window length	1 s = 50 samples
Window step	25 samples (50 % overlap)
Total windows	55 982
Activity-trial groups	15
Per-window normalisation	z-score (mean 0, std 1) per channel

Tab. 2 — Dataset and preprocessing parameters used in this study, as recorded in `reports/metrics.json`.

The pipeline (`privimu.data`) walks the `A_DeviceMotion_data/` directory, parses each `activity_id/sub_K.csv`, applies the windowing of Tab. 2 and yields a feature matrix $X \in \mathbb{R}^{55\,982 \times 60}$ together with subject labels $y \in \{1, \ldots, 24\}$ and group identifiers `activity_trial` used by the cross-validation in §4.

4. Methodology

4.1 *Feature extraction*

For each window of 50 samples and each of the six channels, we compute ten statistics : mean, standard deviation, root-mean-square, minimum, maximum, median, inter-quartile range, skewness, kurtosis and a spectral entropy descriptor over the window's discrete-Fourier magnitude. This yields $10 \times 6 = 60$ features per window. The choice mirrors the canonical HAR feature pipeline of [1, 2] but is re-tasked to identification rather than activity classification.

4.2 *Model*

The primary classifier is a **Random Forest** with 200 trees (scikit-learn defaults otherwise), chosen for its strong baseline behaviour on tabular biometric features, its built-in calibration of probability outputs needed for entropy computation, and its CPU-friendly inference. A 1D-CNN extension exists (`privimu.model_cnn`) but is not the focus of the headline results in §6.

4.3 *Cross-validation strategy*

Random K-fold over individual windows would split a single trial across train and test and produce inflated accuracy. Instead we use a **5-fold GroupKFold** with the group key set to `activity_trial` : each fold holds out a subset of trials, the model is trained on the remaining trials, and predictions are recorded as out-of-fold posteriors over the full $55\,982$ windows. This is the protocol that makes the 39 % top-1 number an honest measurement.

4.4 *Metrics*

We report : top-1 and top-3 accuracy, macro-averaged F1, per-subject F1 (Fig. 3), the 24×24 confusion matrix (Fig. 1), per-window inference latency, and the privacy entropy leakage

$$\Delta H \;=\; \log_2(K) \;-\; H(\mathbf{p}) \;=\; \log_2(24) \;-\; \Big(-\sum_{k=1}^{24} p_k \log_2 p_k\Big),$$

where $\mathbf{p} = (p_1, \ldots, p_{24})$ is the posterior produced by the Random Forest for a given window. ΔH is in bits ; it equals 0 when the posterior is uniform (no information beyond the prior) and $\log_2(24) \approx 4.585$ when the model is fully confident on a single subject.

5. Implementation and reproducibility

The full project is open source at github.com/akiroussama/PrivIMU. The package layout is :

```
src/privimu/
    config.py         # subject ids, default channels
    data.py           # CSV walking, dataset assembly, group keys
    features.py       # 10 stats × 6 channels = 60 features
    model_rf.py       # Random Forest training + persistence
    model_cnn.py      # optional 1D-CNN (extension)
    metrics.py        # entropy, leakage, top-k accuracy, latency
    train.py          # CLI : train + cross-validate + write metrics.json
    evaluate.py       # pretty-prints metrics.json
    viz.py            # confusion matrix, per-subject F1, entropy curve
streamlit_app.py      # interactive privacy lab (§7)
notebooks/PrivIMU_MotionSense_Demo.ipynb
tests/                # 8 pytest unit tests
```

Reproducing the headline result of §6 from a clean clone takes four commands :

```
$ python -m pip install -e ".[dev,app]"
$ python data/download.py --dest data/raw/motionsense
$ python -m privimu.train --data-root data/raw/motionsense \
    --model rf --n-splits 5 --n-estimators 200
$ streamlit run streamlit_app.py
```

SOURCE-OF-TRUTH RULE

Every numerical claim in this report and in the accompanying slides is read from `reports/metrics.json`, the file written by `privimu.train`. If a value is not in that file, it is not in the report.

6. Results

The Random Forest reaches **top-1 accuracy 39.06 %** and **top-3 accuracy 52.80 %** on the held-out windows of the 5-fold GroupKFold, against a uniform random baseline of $1/24 \approx 4.17\,\%$ — a **9.4× lift** on top-1. Macro-F1 is 0.391 ; mean posterior entropy leakage is $\Delta H = 0.700$ bit per window ; per-window inference latency is 42.0 ms on a single CPU. Tab. 3 lists the full configuration and Tab. 4 the headline metrics.

Item	Value
Model	Random Forest, 200 trees
Cross-validation	5-fold GroupKFold
Group key	`activity_trial` (15 unique groups)
Out-of-fold windows evaluated	55 982
Features per window	60
Subjects	24
Random baseline	4.17 % (1 / 24)
Generated	2026-05-13, `reports/metrics.json`

Tab. 3 — Run configuration of the headline experiment.

Metric	Value	Reference
Top-1 accuracy	**39.06 %**	**random : 4.17 %**
Top-3 accuracy	**52.80 %**	**random : 12.50 %**
Macro-F1	0.391	balanced across 24 subjects
Posterior entropy $H(\mathbf{p})$ (mean)	3.885 bit	max : $\log_2(24) \approx 4.585$
Privacy leakage ΔH (mean)	0.700 bit	0 = no leak, 4.585 = full
Latency per window	42.0 ms	CPU, RF.predict_proba
Model size on disk	1 893 MB	200-tree RF, joblib dump

Tab. 4 — Headline metrics. All values read from `reports/metrics.json`.

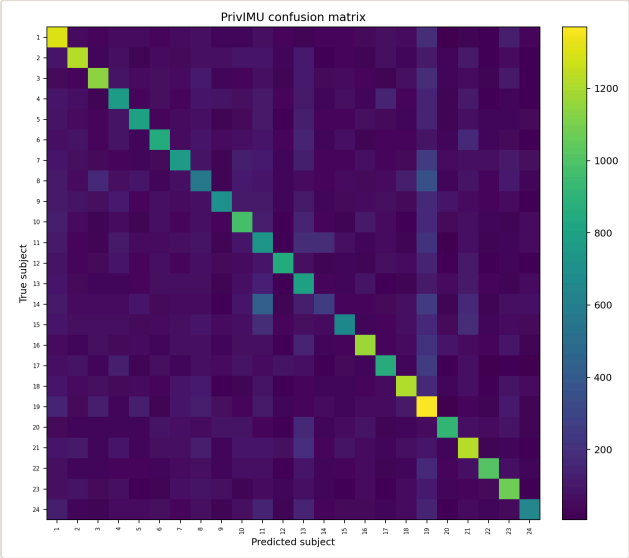

Fig. 1 — 24×24 confusion matrix on the 55 982 out-of-fold windows. The strong yellow-to-green diagonal — especially around subjects 1, 2, 10, 12, 16, 18, 19, 22 — is the visual proof that the model concentrates probability mass on the correct subject. Off-diagonal noise reveals residual confusion between behaviourally close users.

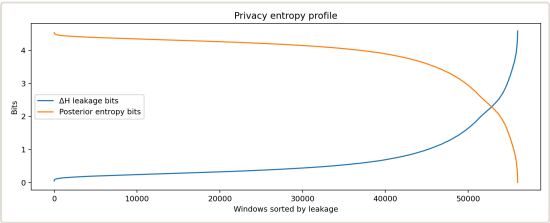

Fig. 2 — Privacy entropy profile over the 55 982 windows sorted by leakage. The right tail captures the $\sim 5\,\%$ of windows whose posterior collapses on a single subject — the dominant driver of identification risk.

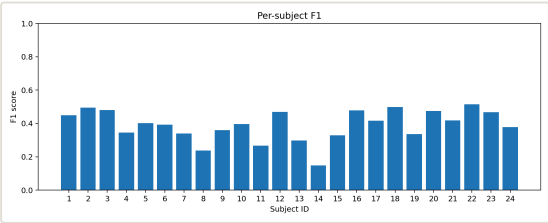

Fig. 3 — Per-subject F1. Identifiability is not uniform : $\mathrm{F1} \approx 0.51$ for the most identifiable subject (22), $\mathrm{F1} \approx 0.15$ for the least (14). Some users are intrinsically more exposed.

7. Interactive privacy lab

To make the result tangible — and to give the audience something to inspect rather than read — we ship a Streamlit application (`streamlit_app.py`) that wraps the trained model in a six-stage interactive flow :

1. **Source.** The user picks one of the `examples/wlk_7/sub_*.csv` traces shipped in the repository or uploads their own MotionSense-like CSV.
2. **Signal plot.** Plotly traces of the six IMU channels.
3. **Top-3 ranking.** Window-aggregated posterior over the 24 enrolled subjects, presented as a ranked bar chart with confidence values.
4. **Identity-lock replay.** Window-by-window posterior, with a confidence curve and a "lock-time" marker indicating when the top-1 hypothesis stabilises.
5. **Defence sweep.** A slider for the noise standard deviation σ (relative to the signal standard deviation) injects Gaussian noise on the raw channels before feature extraction. The app plots the trade-off between top-1 confidence, normalised leakage and a signal-preservation proxy.
6. **Evidence card.** One-click export of a JSON + Markdown bundle (source CSV, top-3, ΔH, σ, identity-lock window, timestamp) for inclusion in a security review.

The application can run without the trained model in a degraded UI-only mode, useful for design reviews. That mode is clearly marked and is never used for the numerical claims in this report.

8. Discussion — risks, limitations and mitigations

8.1 *What 39 % top-1 actually means*

A 39 % top-1 accuracy on 24 subjects with an off-the-shelf Random Forest is not a real-world identification system. It is, however, sufficient to establish two facts. First, the inertial signal carries a behavioural signature that survives the removal of every explicit identifier — the lift over the random baseline is almost an order of magnitude. Second, the leakage is not uniform : Fig. 3 shows that a sub-population is substantially more identifiable than the average, and Fig. 2 shows that a small fraction of windows leak the bulk of the information. Treating "anonymous IMU" as a uniformly low-risk data type is therefore not defensible.

8.2 *Limitations*

- **Closed-set.** Evaluation assumes the target user is one of 24 known candidates. Open-set detection (rejecting unknown subjects) is not

addressed.

- **Population.** 24 subjects is small. The honest interpretation is that the result establishes existence of leakage, not its scaling law with population size.
- **Single device class.** All recordings come from an iPhone 6s in the front pocket. Cross-device or cross-placement transfer is out of scope.
- **Single dataset.** MotionSense was collected in controlled lab conditions. Behaviour in the wild may carry more noise and less stable signatures.
- **Adversarial robustness.** The defence sweep uses Gaussian noise as a pedagogical baseline ; it is not a certified differentially-private mechanism.
- **Static model.** No online adaptation, no temporal model across windows beyond posterior aggregation.

8.3 *Privacy-by-design mitigations*

The defence stack a deployable IoT product should consider, in increasing implementation cost :

1. **Data minimisation.** Lower the sample rate to the minimum the use-case requires. Drop channels that the application logic does not consume.
2. **On-device inference.** Compute the application output (steps, falls, gestures) locally and never ship the raw window to the cloud.
3. **Federated learning [9].** When centralised aggregation is required, train without collecting raw traces : only model deltas are exchanged.
4. **Differential privacy [10].** When statistics on motion are released, use calibrated noise on the aggregates with a documented privacy budget.
5. **Feature-level sharing.** Release coarse statistics (per-day step count, per-week stride variability) rather than raw 50 Hz windows.
6. **Consent and transparency.** Treat IMU streams as personal data in privacy notices, with granular opt-in and short retention.

> **ETHICAL SCOPE**
>
> This work uses a public, peer-reviewed dataset (MotionSense, 24 consenting subjects). No model trained here is shipped against real users. The intent is **defensive** : convince engineers and architects to handle motion data with the same care they apply to faces and voices.

9. Conclusion

PrivIMU shows, on a public 24-subject benchmark, that anonymous-looking IMU traces carry enough behavioural signature to lift a closed-set identification

accuracy from 4 % (random) to 39 % (Random Forest) and to leak 0.7 bit of identity per 1-second window on average. The result is reproducible from a clean clone in four commands ; it ships with a Streamlit lab that lets a non-specialist witness the attack and tune a first defence ; and it is paired with an explicit list of limits and a privacy-by-design mitigation checklist. The wider message for the FF6 IoT setting is that motion data should be treated as personal data — not because every IMU trace identifies every user, but because the signal carries identity in a way that grows with population, model and data scale, and the privacy posture must be set *before* that scale arrives.

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